



YOUR TRUSTED HARDWARE ISN'T: WHY SILICON BELONGS IN THE THREAT MODEL



Except where separate rights are noted

OWASP Ottawa Chapter Meetup

↘ uOttawa, Nov 12, 2025, public



\$ WHOAMI



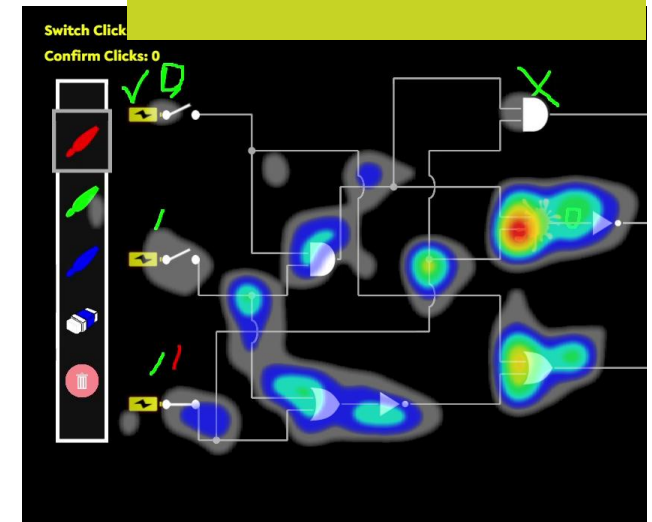
René Walendy

Embedded Security Researcher

➤ *Max Planck Institute for Security and Privacy,
Bochum, Germany*

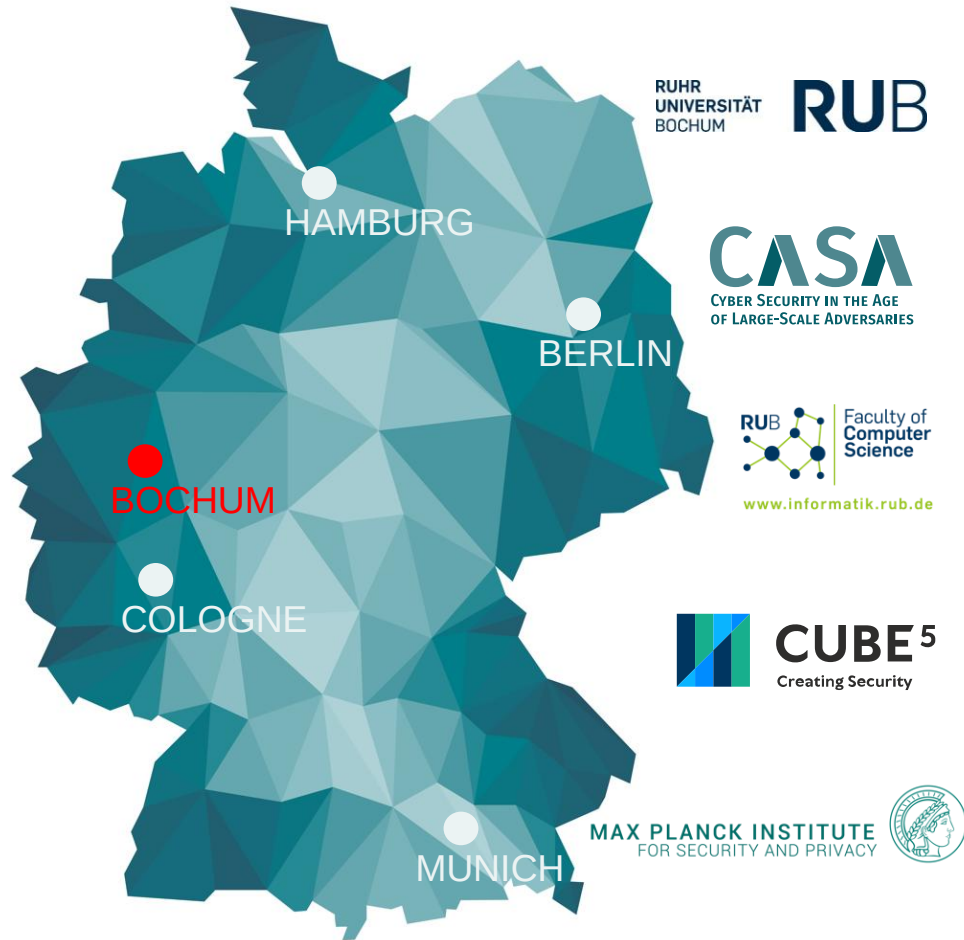
- Interdisciplinary research
@ MPI-SP
- Teaching reverse engineering
@ Ruhr University Bochum

HUMAN FACTORS in hardware security





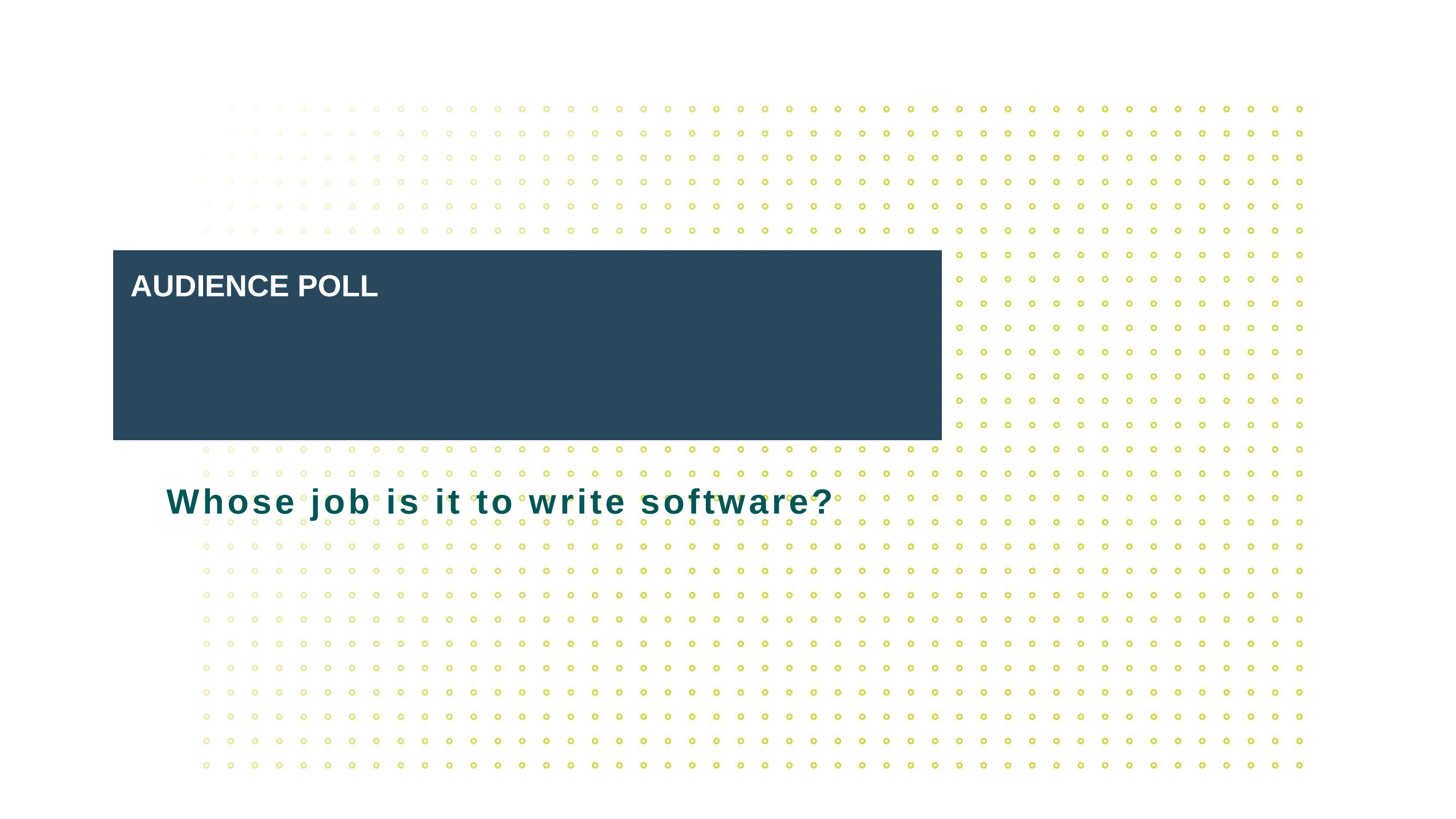
HARDWARE & IT SECURITY RESEARCH IN BOCHUM



300+ Researchers

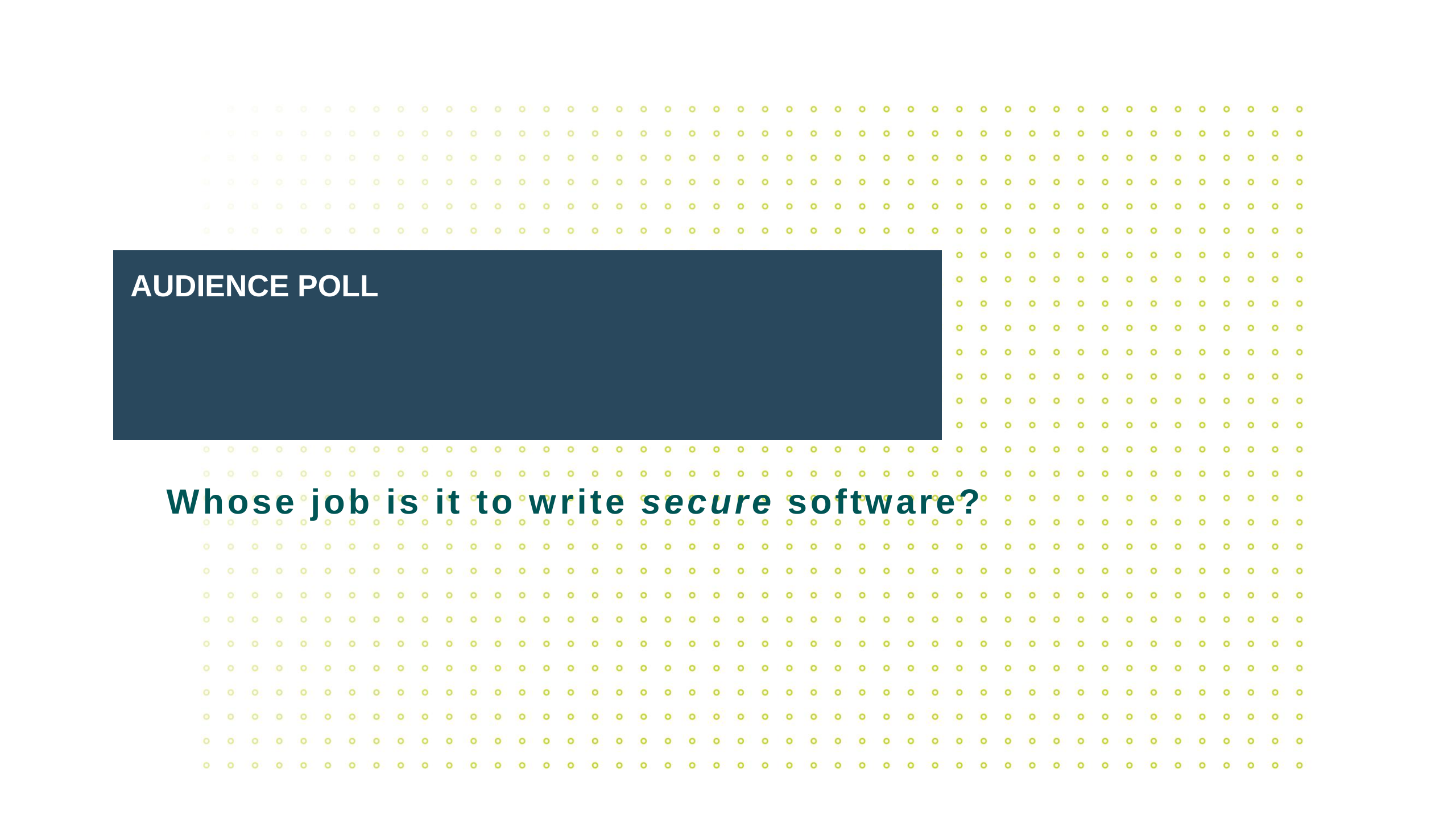
1000+ Cybersecurity Students





AUDIENCE POLL

Whose job is it to write software?



AUDIENCE POLL

Whose job is it to write *secure* software?



AUDIENCE POLL

Who is considering *hardware* security?

AUDIENCE POLL

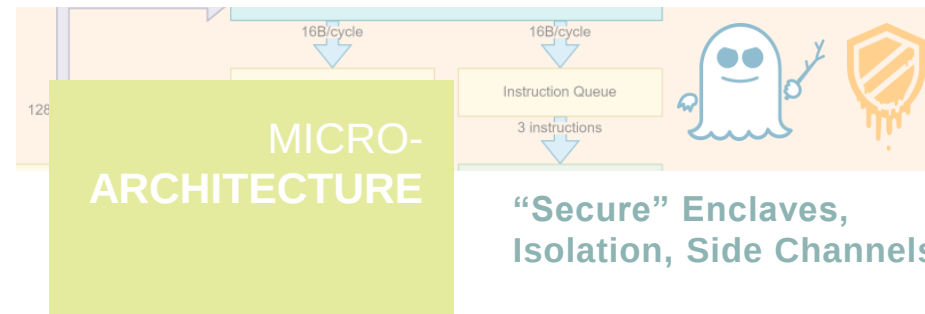
Who here trusts the hardware vendor of your servers?

... BUT SHOULD YOU?



SCOPE OF THIS TALK

Layers of Security



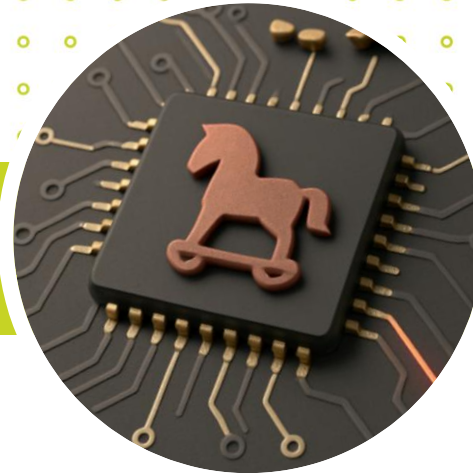
Wafer image by 2x910 on Wikimedia



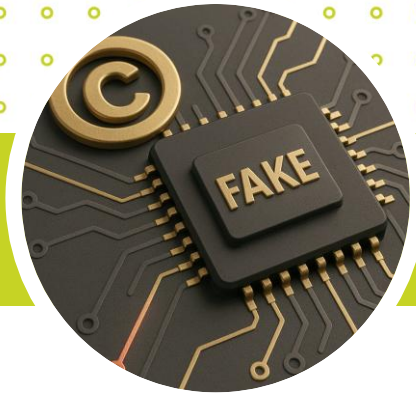
THREAT MODELS



Cryptographic
secret extraction



**Hardware
Trojans**



Intellectual property infringement
& Counterfeiting



HARDWARE SUPPLY CHAINS

Are complex and opaque





WHEN THE SUPPLY CHAIN MESSES UP ...

... You get F00Fed

Say the magic words ...

F0 0F C7 C8

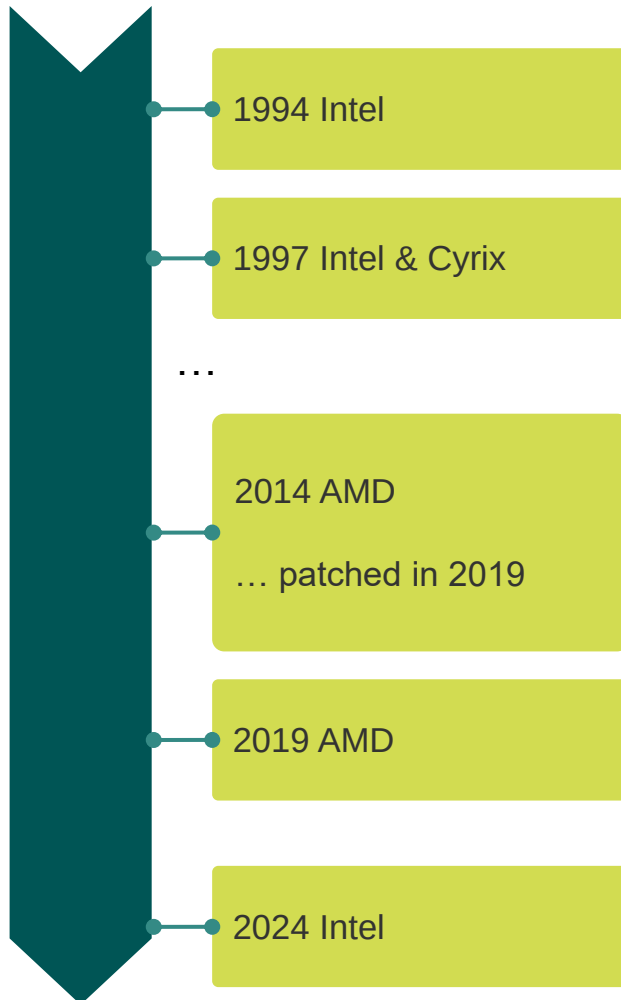
lock cmpxchg8b eax

... for a hardware denial of service from userspace!





... AGAIN AND AGAIN AND AGAIN



Hardware bugs *do* silently slip through validation testing!



THESE BUGS WERE ACCIDENTAL
... but they can be weaponized!

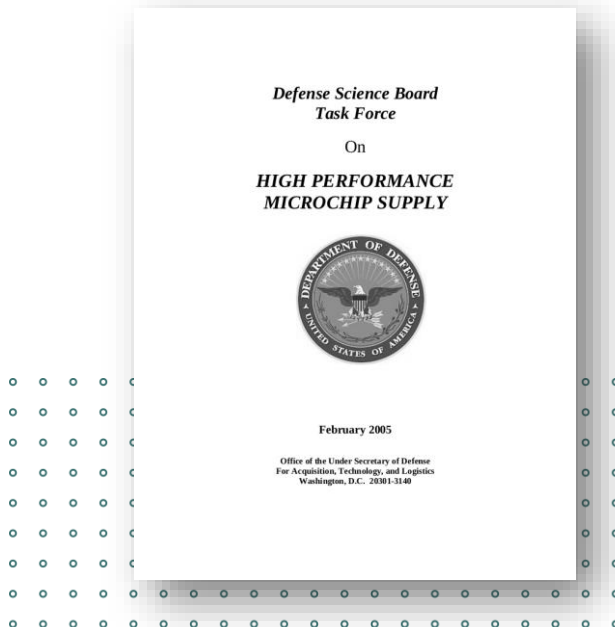


HARDWARE TROJANS AND THE SCIENTIFIC COMMUNITY

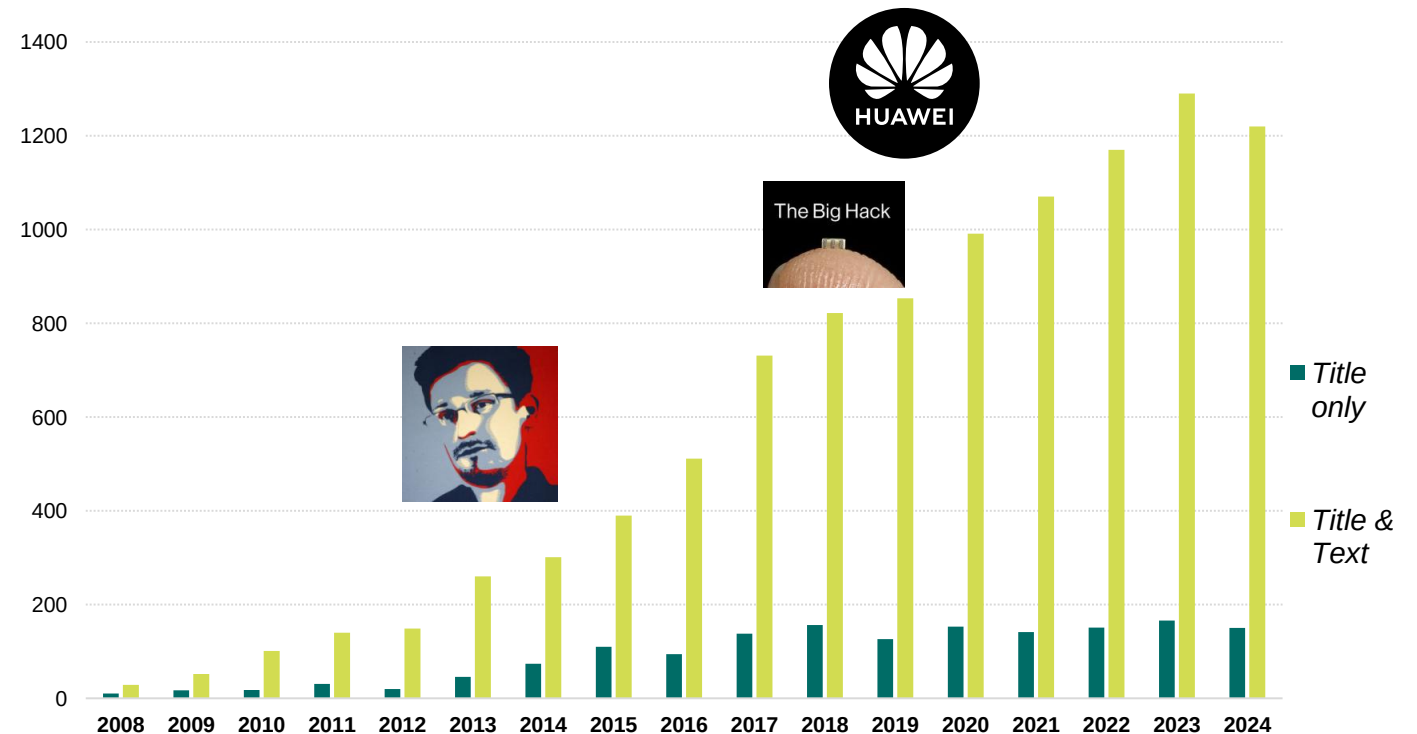
Publications 2007 .. 2024

DoD 2005 Report

US Military sketched out
about hardware supply



PUBLICATIONS ON “HARDWARE TROJAN(S)” IN GIVEN YEAR



Source: Google Scholar, as of Nov 02, 2025



A RANDOM EXAMPLE

for trusting your hardware

```
>>> import secrets  
>>> secrets.randbelow(0, 0xFFFF)  
  
# ???  
  
61455
```

OS magic



Picture by Ashley Pomeroy on Wikimedia

Proof of Security

Let k be a fresh and uniformly distributed random key, then ...





HARDWARE TROJANS

Can be hidden in plain sight

18 billion
transistors

< 10 nanometers
in diameter each



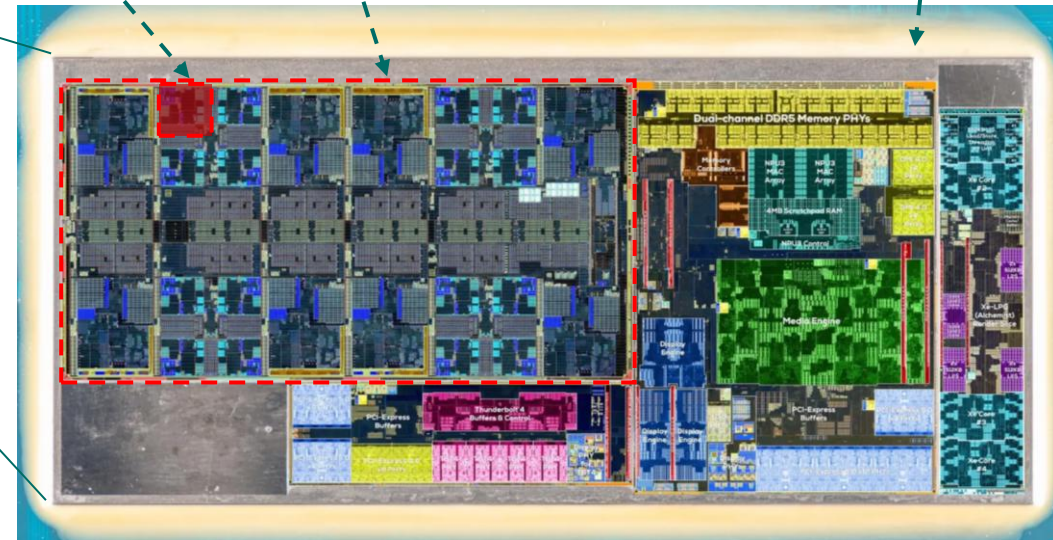
© Intel

One CPU core

Compute logic

„Everything else“

(the RNG is somewhere in here)
(it's about this size: .)



© x.com/highyieldYT

approx. 26mm

- Step 1: Manipulate random number generator
- Step 2: Don't get caught
- Step 3: ????
- **Step 4: Profit!!**



LET'S DESIGN A HARDWARE TROJAN

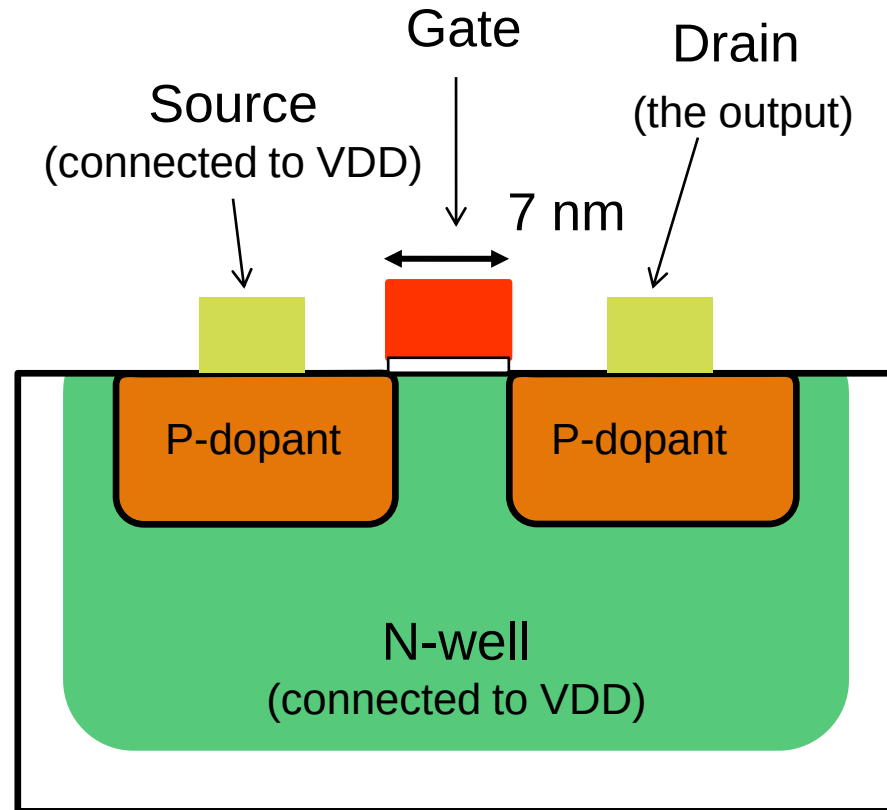
... for fun and profit!

Becker et al. "Stealthy Dopant-Level Hardware Trojans: Extended Version" (2014)

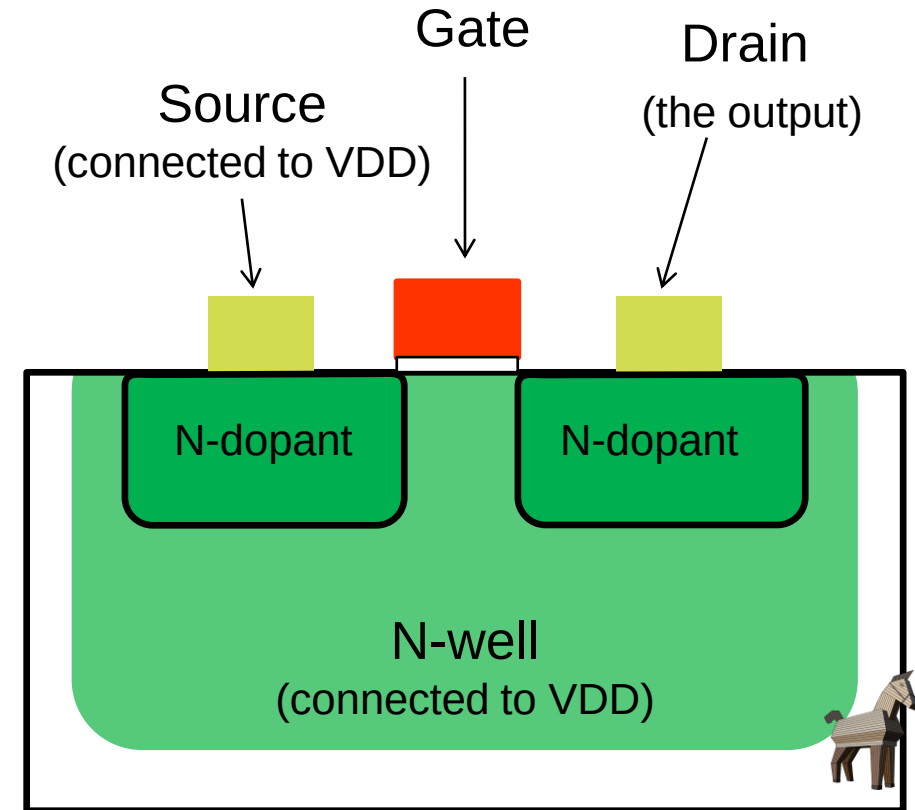


TROJAN BUILDING BLOCK

“Stuck” PMOS Transistor



Unmodified PMOS transistor

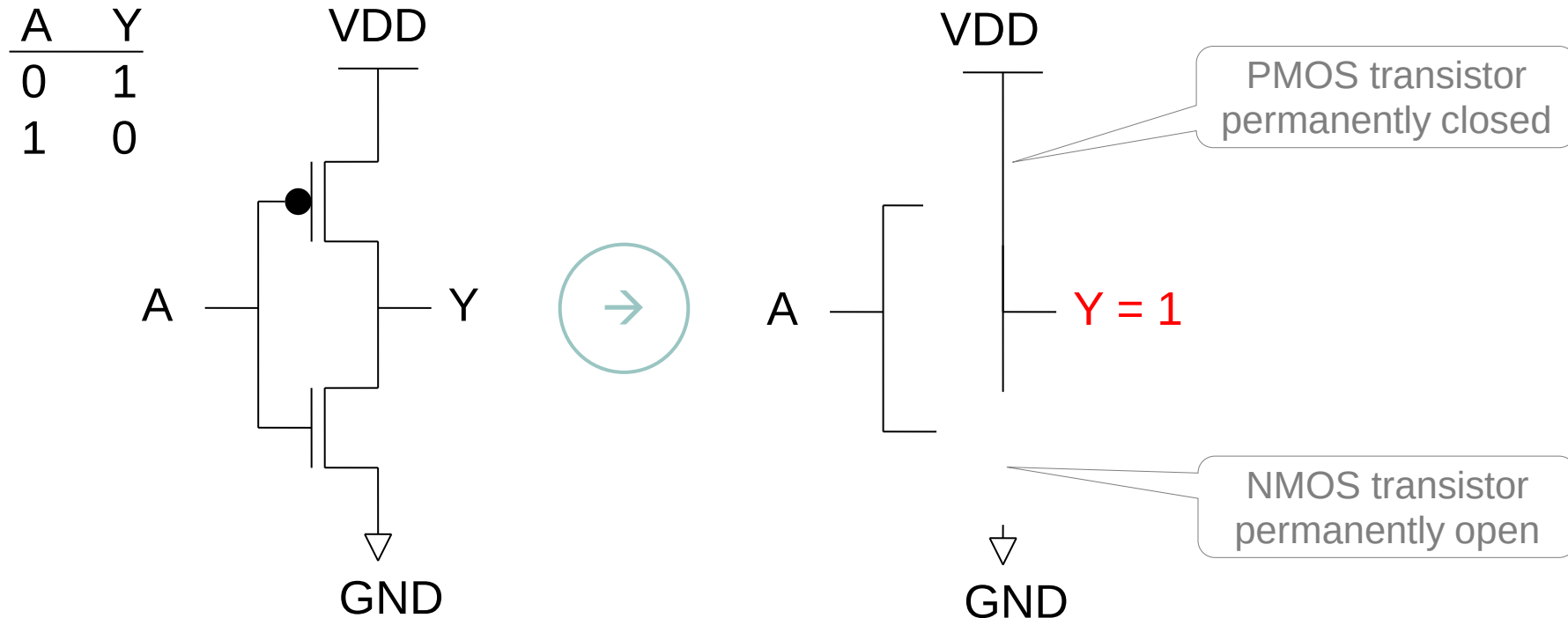


Trojan transistor “always 1”



TROJAN BUILDING BLOCK

“Stuck” PMOS Transistor



Looks inconspicuous ...

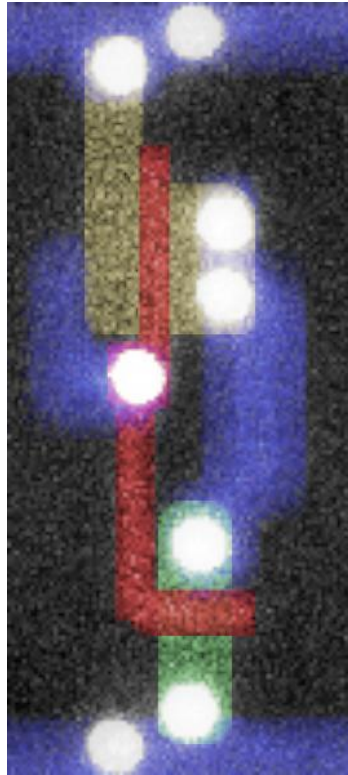
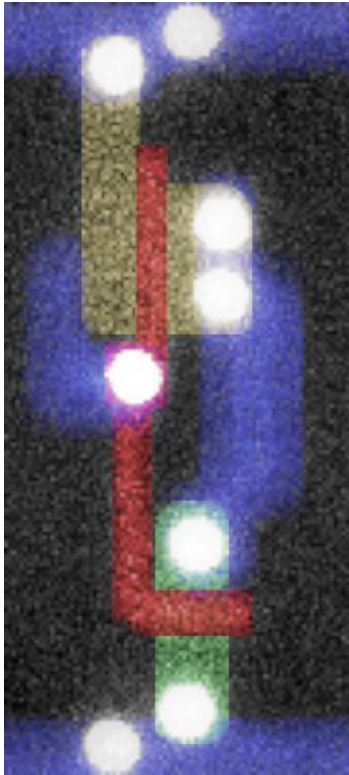
... but behaves maliciously



IS THIS DETECTABLE?

Let's try!

Which one has the Trojan?



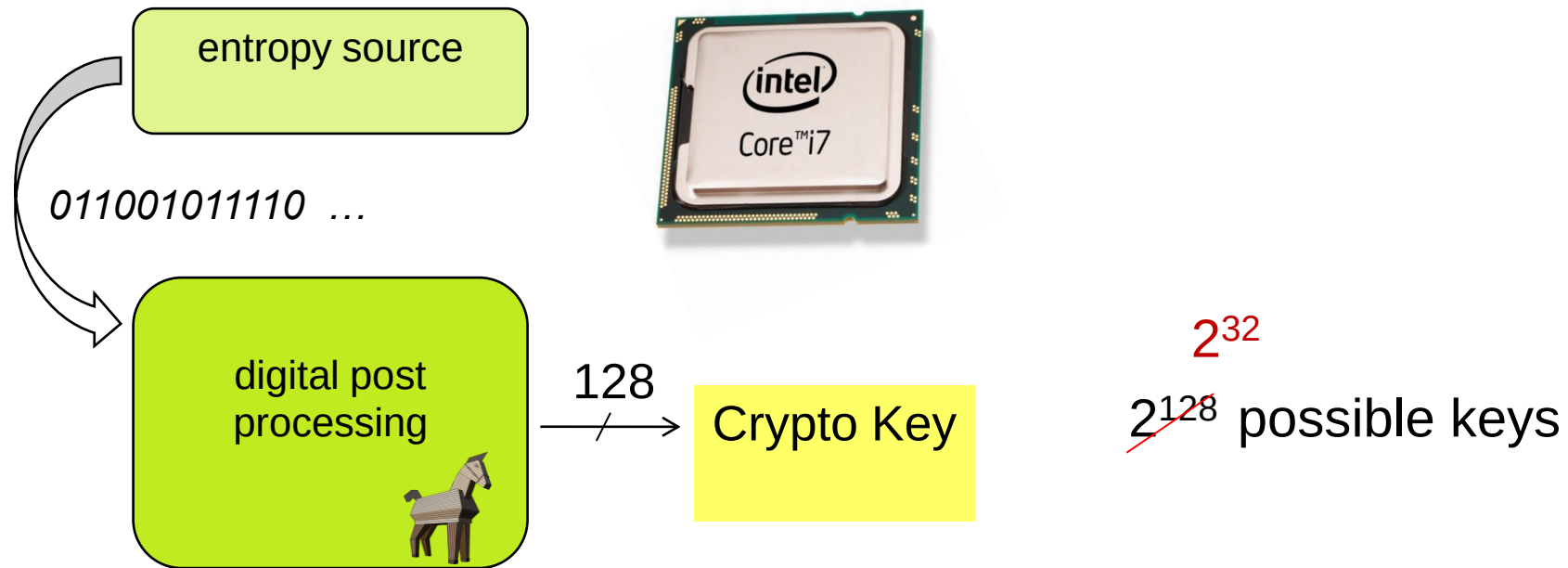
Dopant hacking is difficult to detect using microscopy inspection!

** chemical sample preparation techniques for dopant detection exist, but they are slow and expensive*



RANDOM NUMBER GENERATOR TROJAN

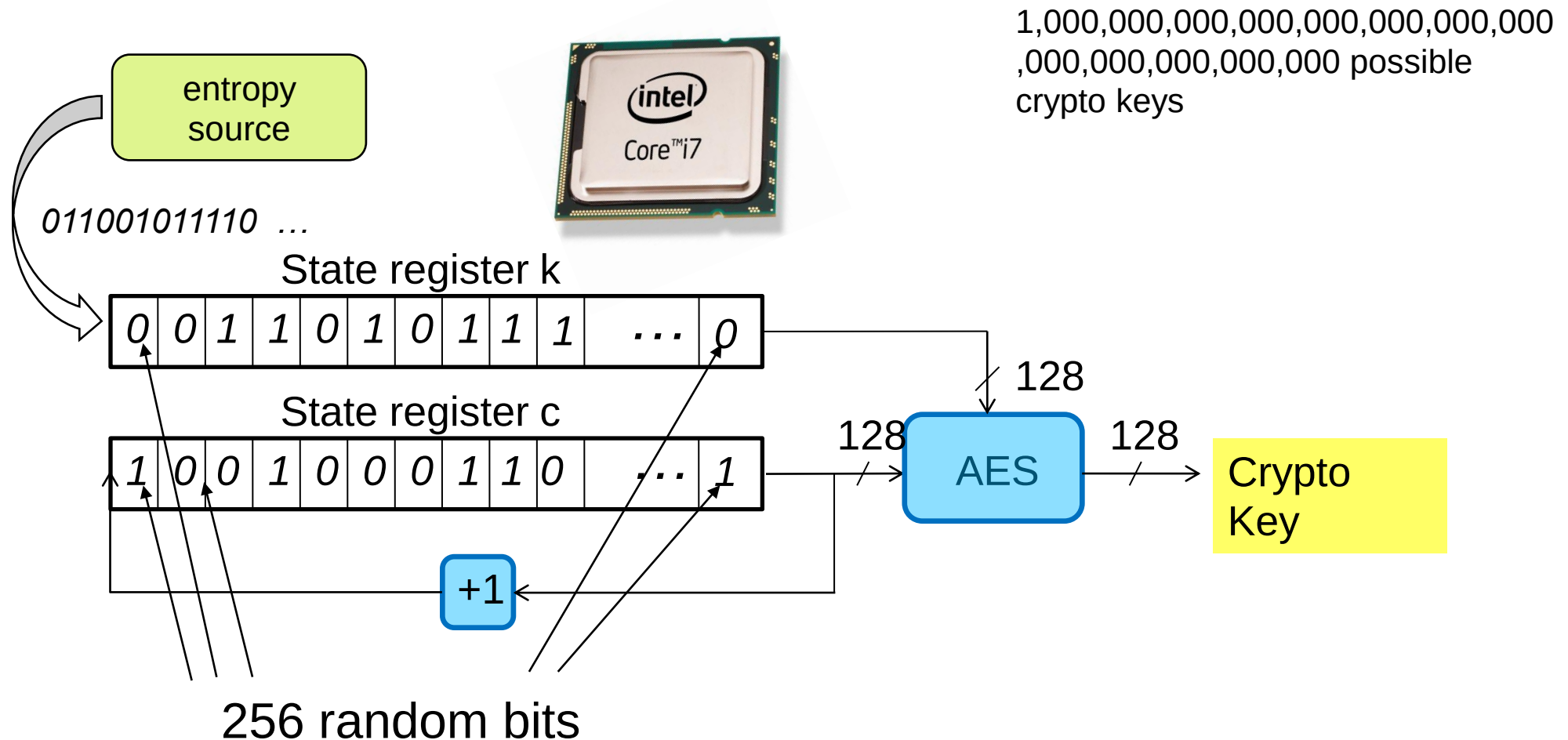
Concept in a Nutshell





RANDOM NUMBER GENERATOR TROJAN

Inner Workings



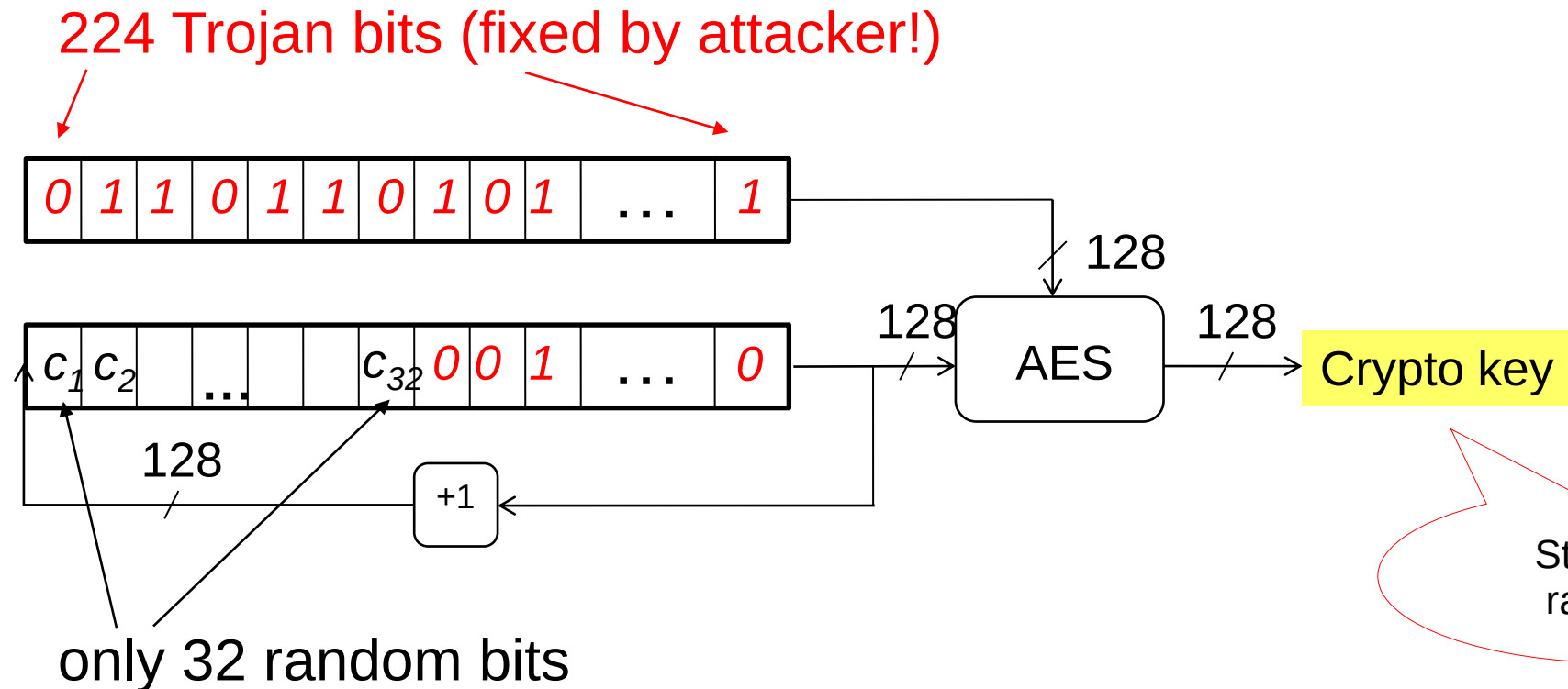


RANDOM NUMBER GENERATOR TROJAN

Inner Workings

Testing all keys:
few seconds

1,000,000,000,000,000,000,000,000,000
,000,000,000,000,000 possible
crypto keys





HARDWARE PATCHES

Are practically impossible



**Manipulated hardware can subvert
software defenses.**

Patching deployed hardware is really difficult!



TROJAN DESIGN: TAKEAWAYS

SECURITY GUARANTEES OFTEN ASSUME HONEST HARDWARE

Cryptographic proofs depend on true randomness, generated by RNGs on the CPU.

HARDWARE TROJANS CAN UNDERMINE HARDWARE INTEGRITY

Small and deliberate circuit manipulations can be stealthy and effectively unpatchable.

THE ATTACK VECTOR LIVES OUTSIDE THE THREAT MODEL

This is how „provably secure“ systems get owned!



HOW TO DEFEND

... against silicon-level threats?

Puschner et al. “Red Team vs. Blue Team: A Real-World Hardware Trojan Detection Case Study Across Four Modern CMOS Technology Generations” (2023)

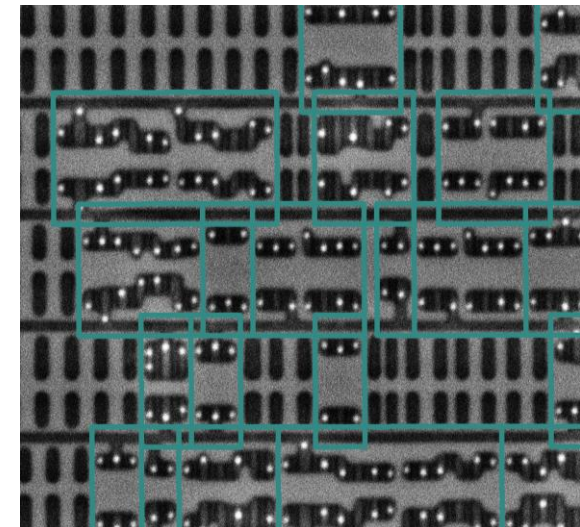


FINDING TROJANS

Some insights into current research



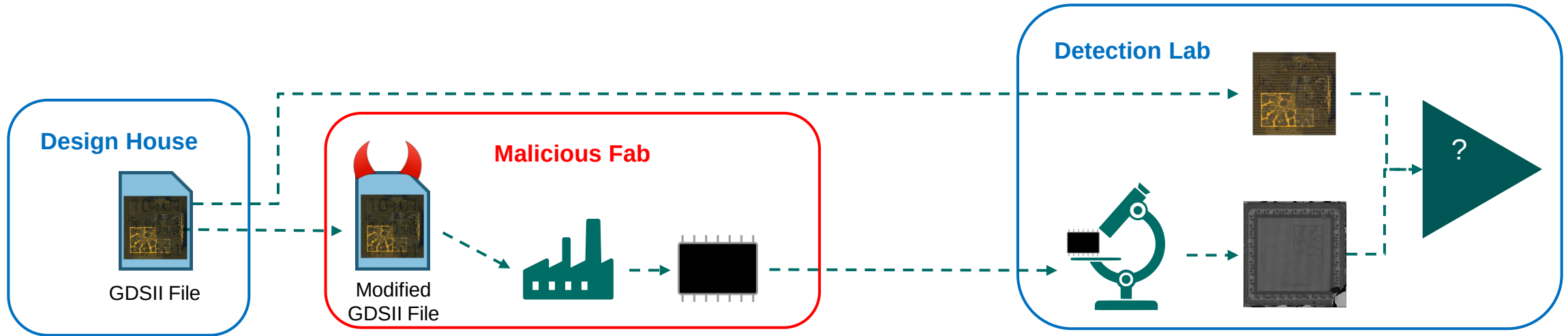
The State-of-the-Art in IC Reverse Engineering, R. Torrance and D. James, 2009





THREAT MODEL & DETECTION

Of Silicon-Level Hardware Trojans



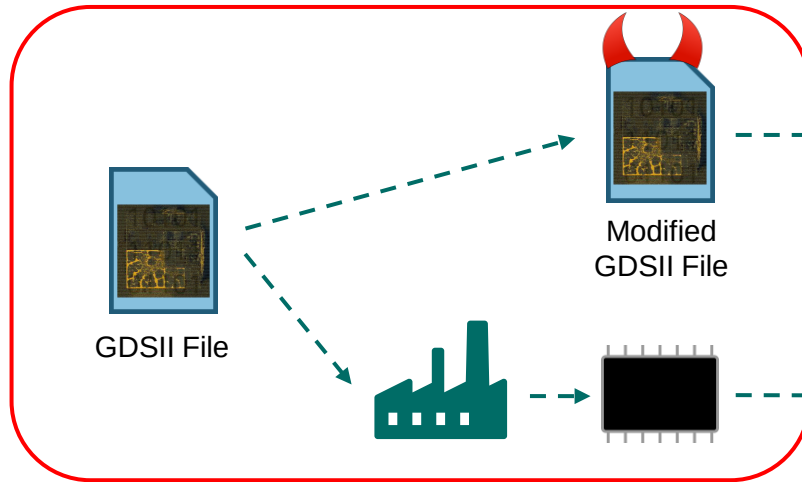
- Fab-less design house sends design file to fab
- Malicious fab edits the design before production (replaces cells)
- Detection lab compares original design file to the manufactured chip
- **Other steps are “Trojan free”**



OUR APPROACH: RED TEAM VS. BLUE TEAM

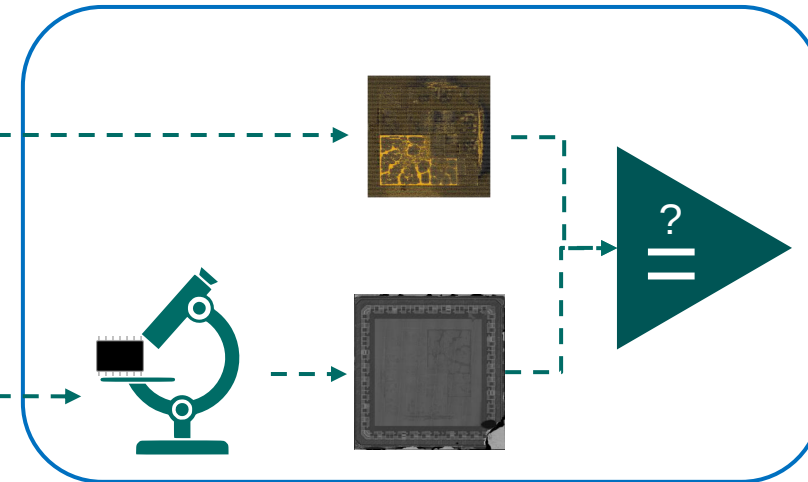
Purpose: minimize research bias

RED TEAM (resembles malicious third party)



→ creates delta between chips and design files

BLUE TEAM (resembles analysis lab)

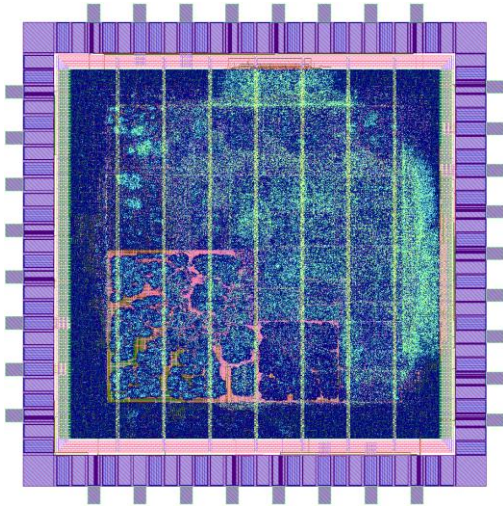


→ receives chips & design files from **RED TEAM**

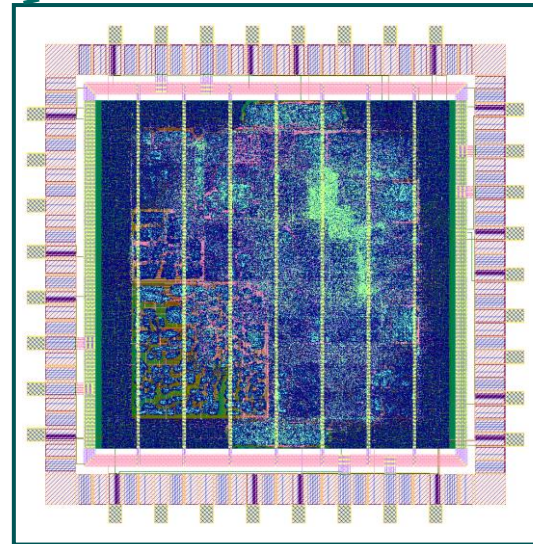
→ tries to find the differences



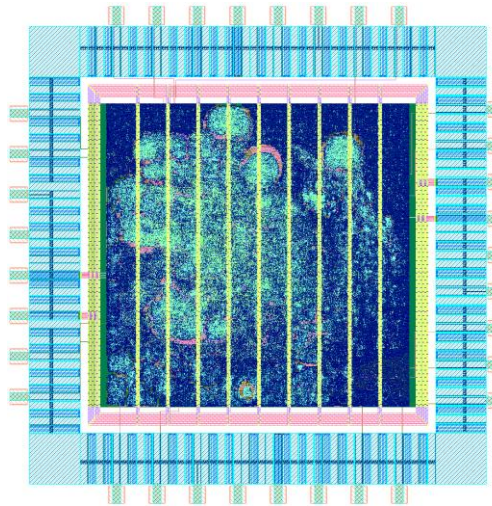
FOUR TARGET CHIPS



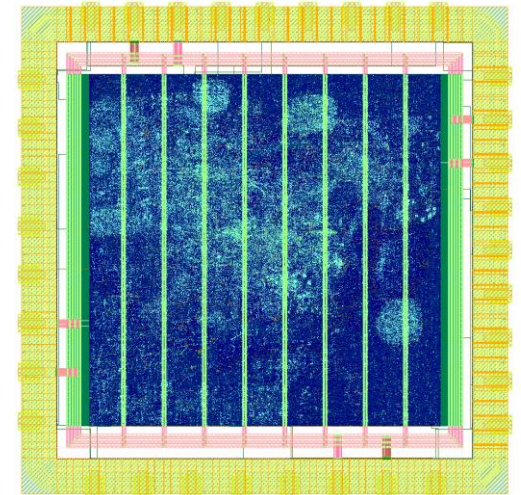
a) 90nm IC



b) 65nm IC



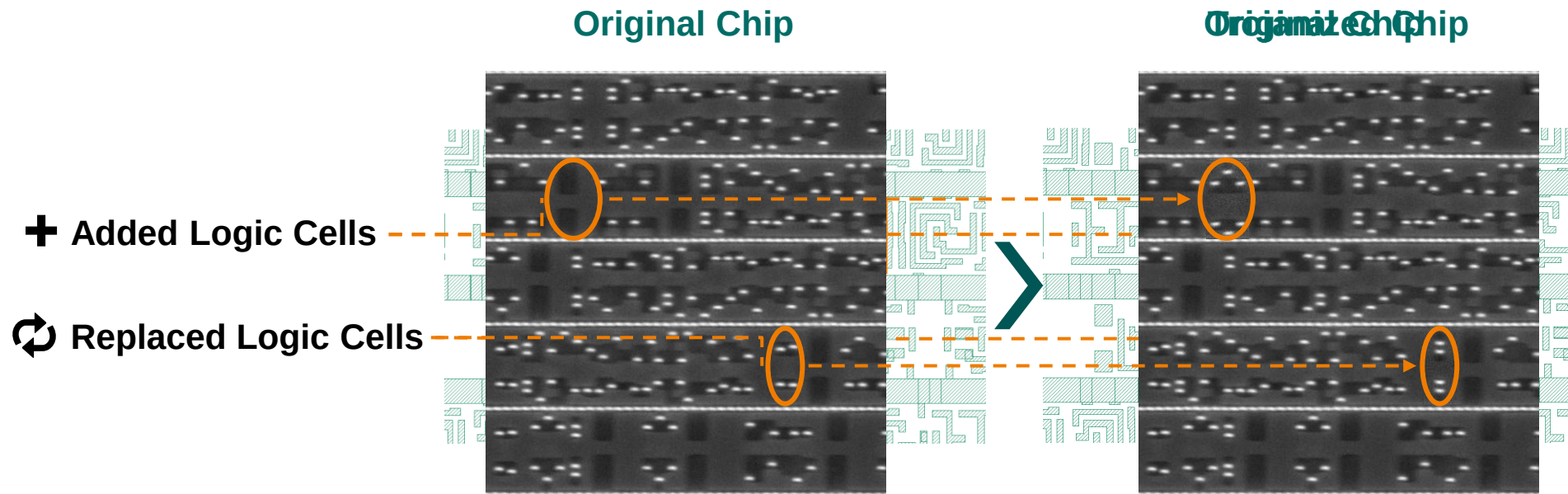
c) 40nm IC



d) 28nm IC



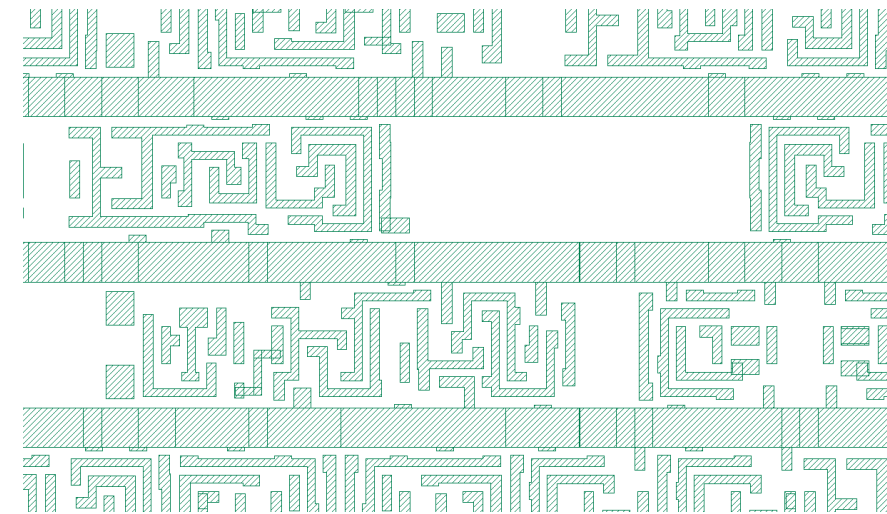
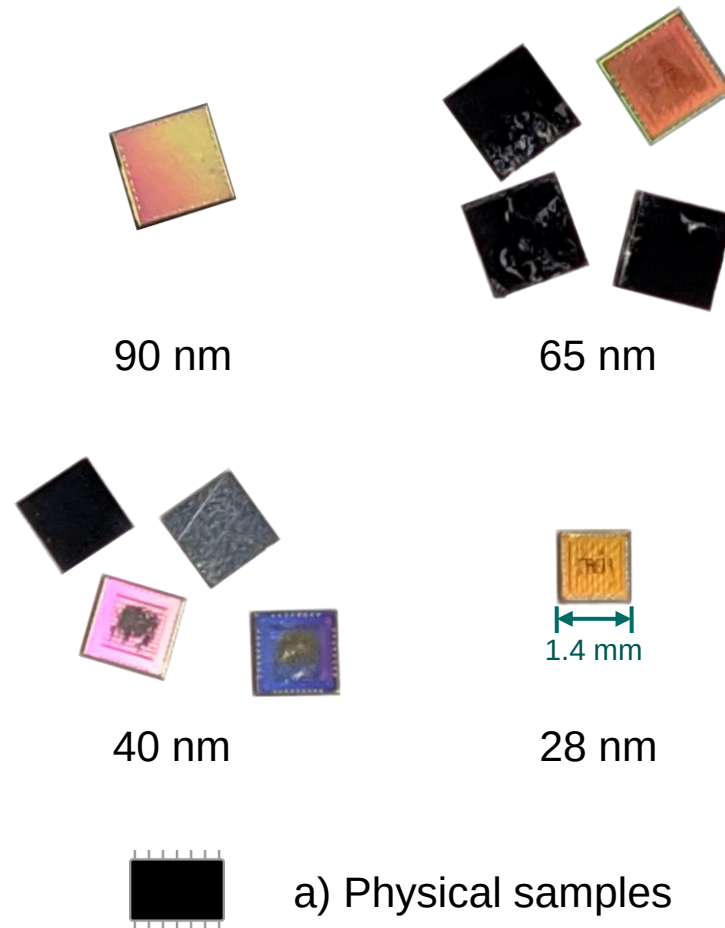
RED TEAM: EMULATE INSERTED TROJAN



- **RED TEAM** introduces ten modifications per chip at random positions:
 - 6 x + Added logic cells
 - 4 x ↻ Replaced logic cells
- 40 of 3,410,580 total cells → 0.001 % modified

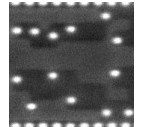
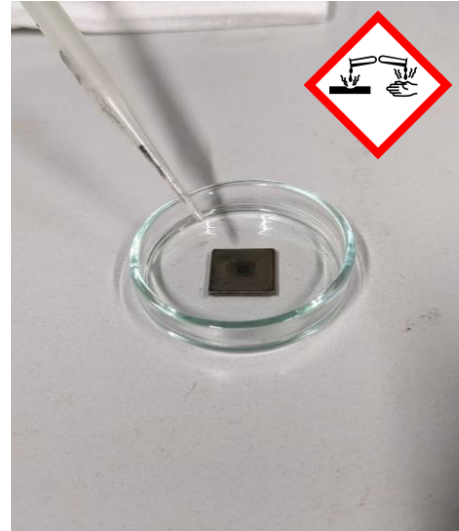
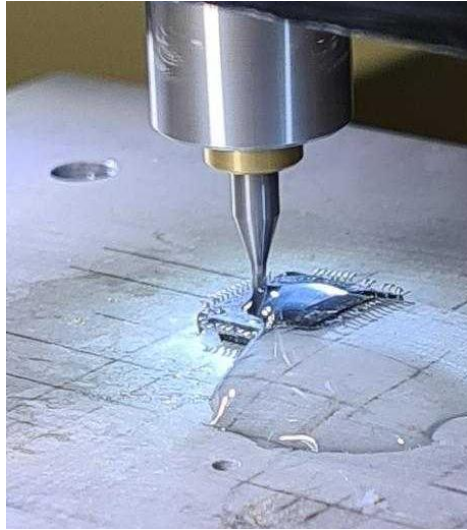


BLUE TEAM RECEIVES...



b) GDS-II Design

SAMPLE PREPARATION & IMAGING



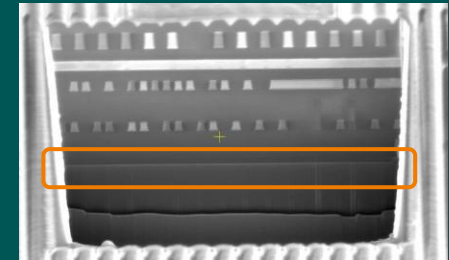
1. CNC milling

2. Choline hydroxide etching

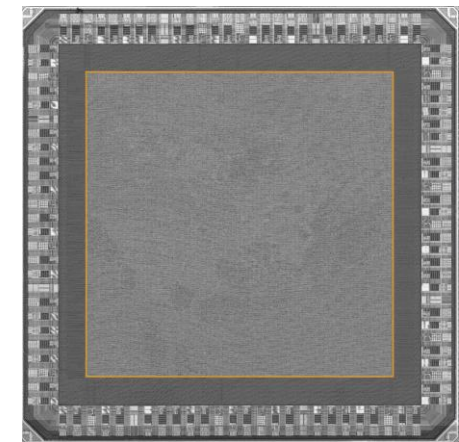
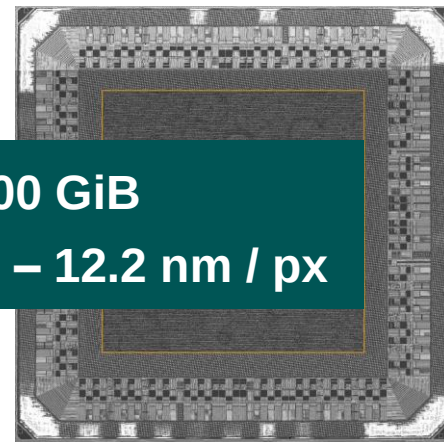
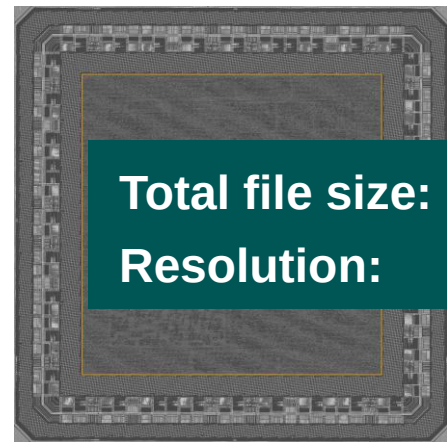
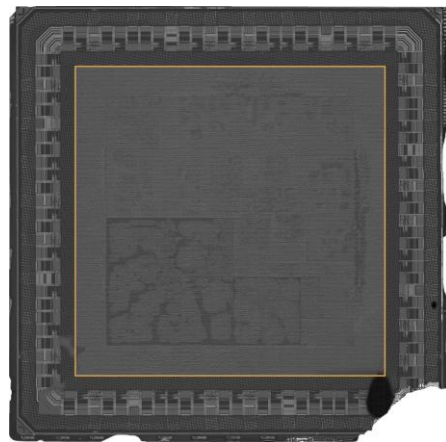
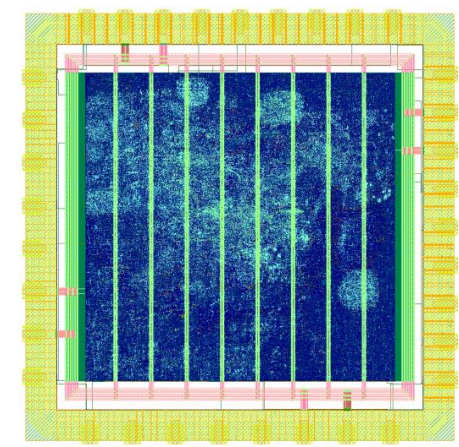
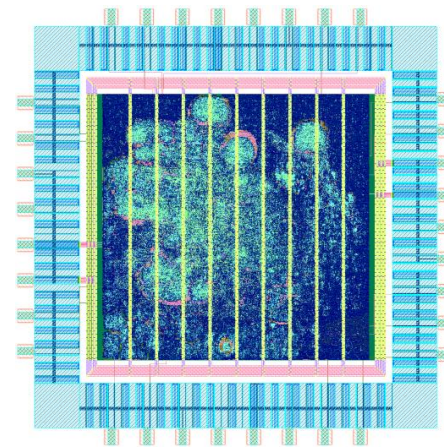
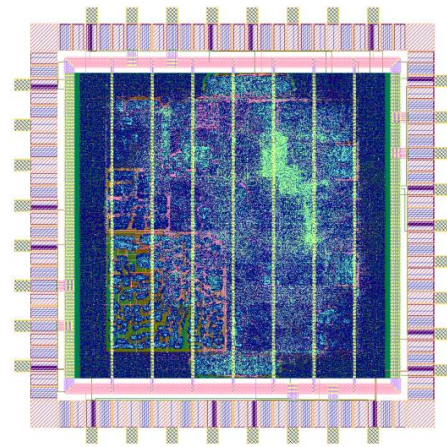
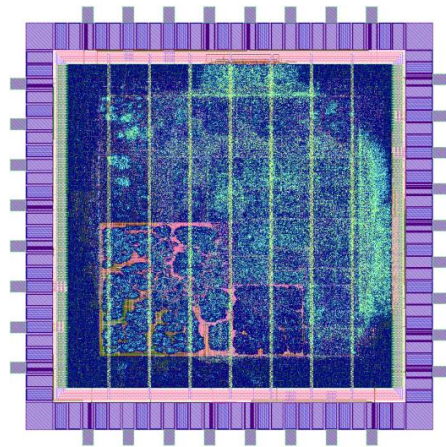
3. Scanning electron microscope (SEM)

Cross-section Side View

- Cell layout: Bottommost layer
- Removing silicon from the bottom
- Imaging the back side



LAYOUT VS. SEM BACKSIDE IMAGE



Total file size: ~300 GiB
Resolution: 4.8 – 12.2 nm / px

a) 90nm IC

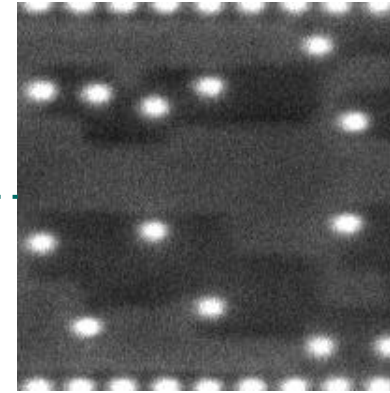
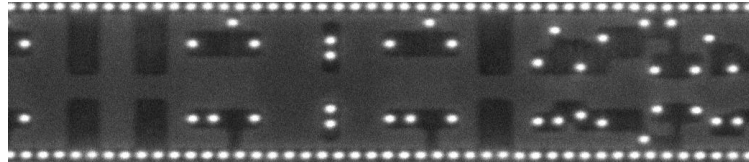
b) 65nm IC

c) 40nm IC

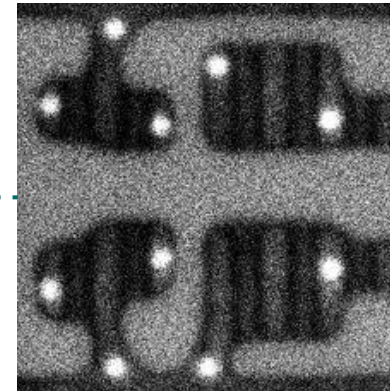
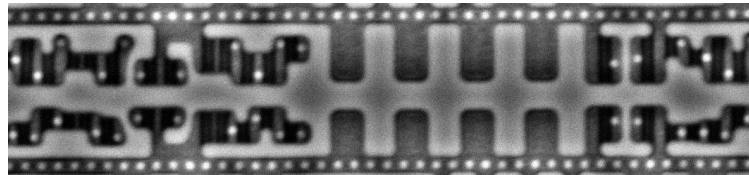
d) 28nm IC

SEM IMAGES

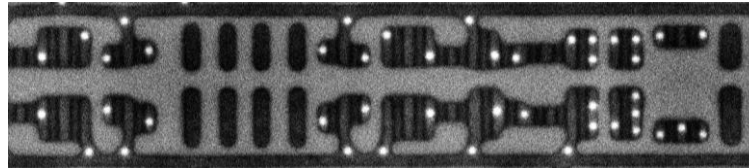
a) 90nm IC



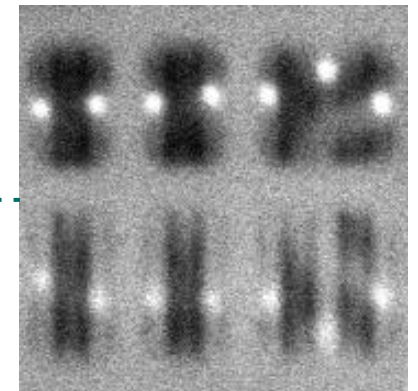
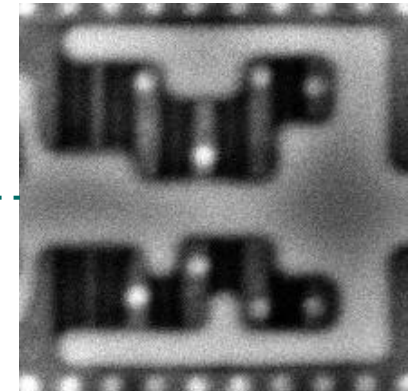
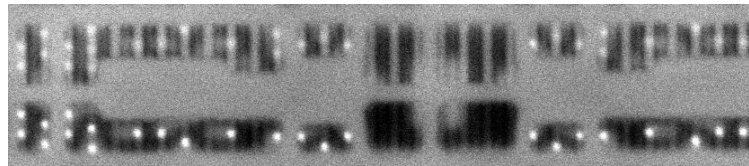
b) 65nm IC



c) 40nm IC

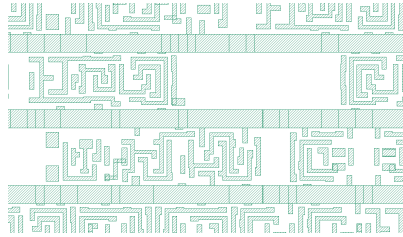


d) 28nm IC





BLUE TEAM DETECTS MANIPULATIONS



Alignment 1

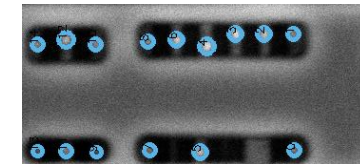
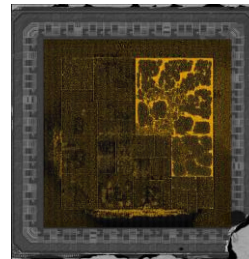
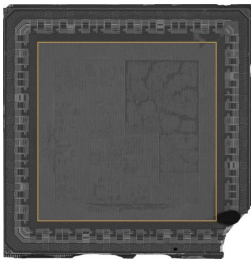
Gather cell coordinates
from GDSII design file

Alignment 2

Gather tile coordinates
from stitched SEM images

Align Design & Images
Apply transformation

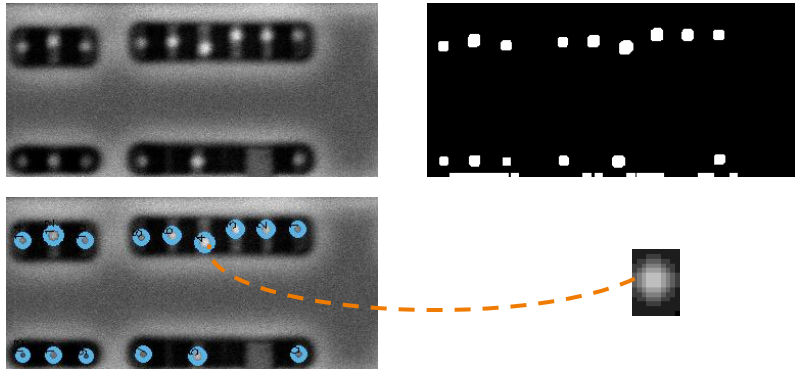
Detect Manipulations
Feature detection /
Comparison algorithms



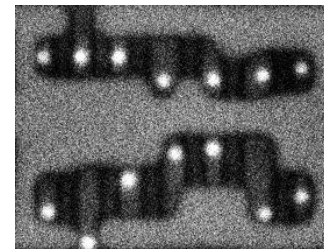


DECISION ALGORITHMS

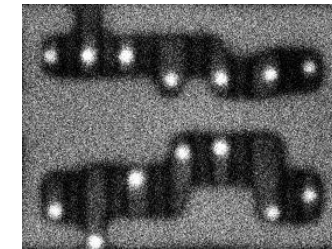
+ Additional Cells: Via Detection



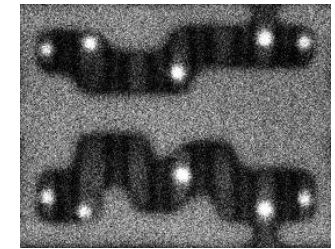
↻ Replaced Cells: Template Matching



Reference model



Same label



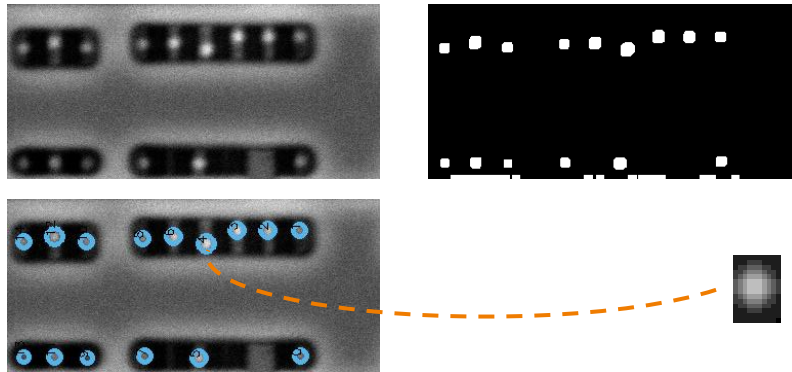
Same label



FINDING LOGIC CELLS WHERE FILLER CELLS ARE EXPECTED

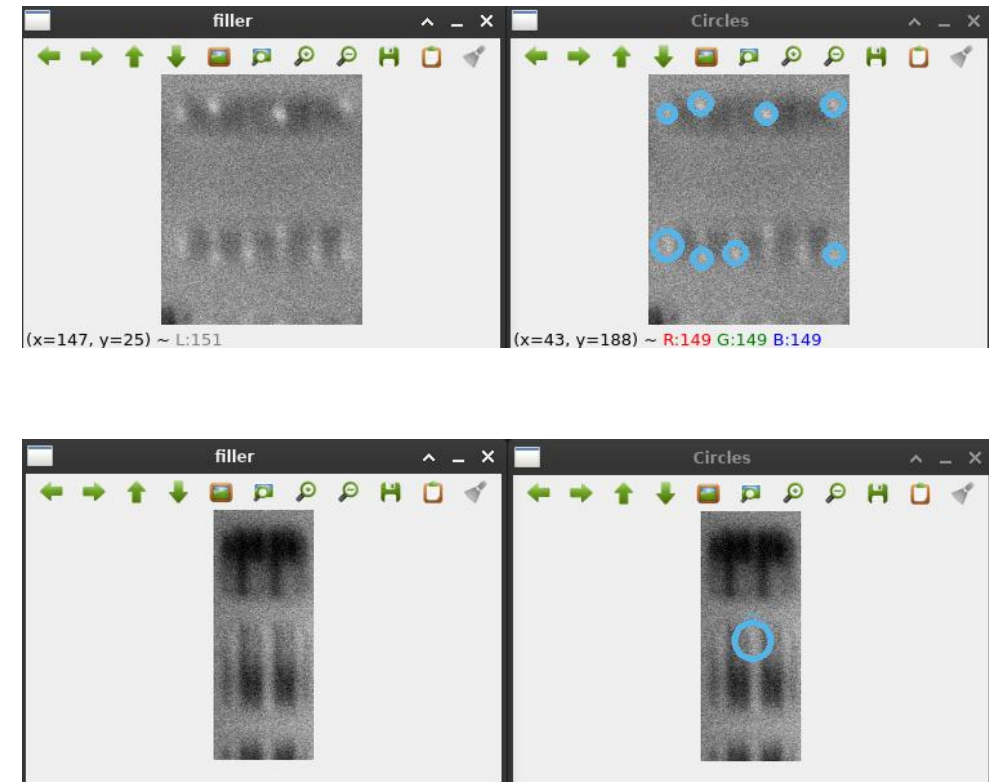
Cells contain vias, let's detect them

- Approach:
 - Suppress noise, threshold, find spots of defined size
 - Build a gradient and correlate (corr > x = via)



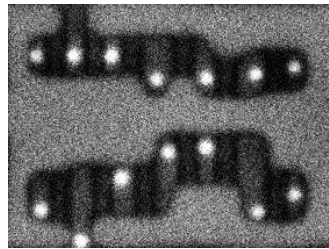
In the end, depends on image quality and parameters

→ some false positives

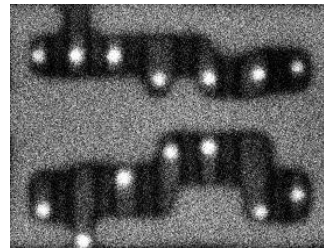


FINDING CELL REPLACEMENTS

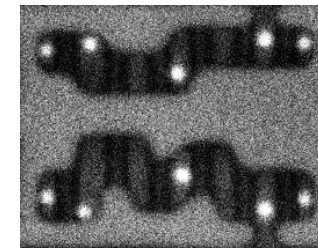
- Q: “Is the cell in question the one it is labeled, or was it replaced by another cell?” (e.g., NAND → NOR)



Golden Model / Template



Other Cell labeled same



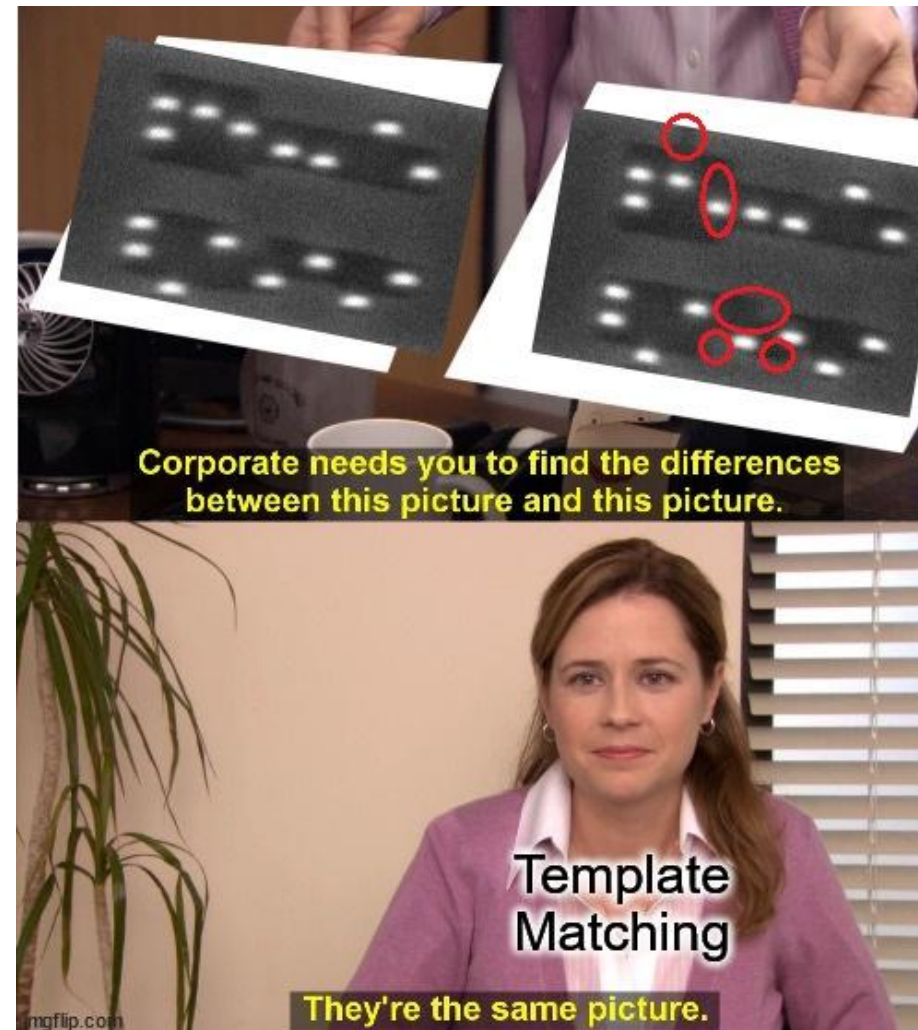
Other Cell labeled same



→ Template Matching

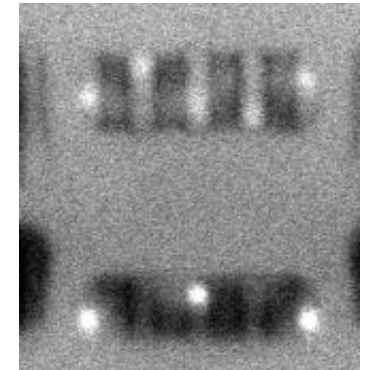
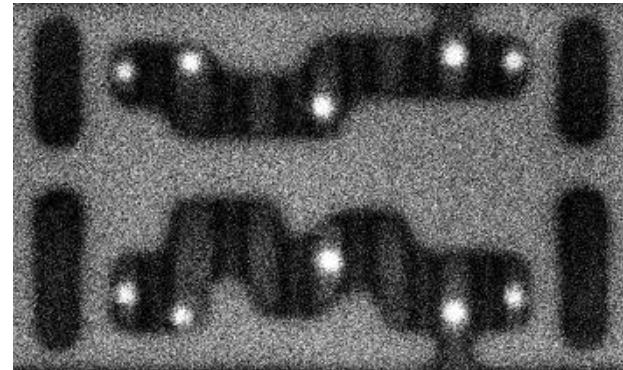
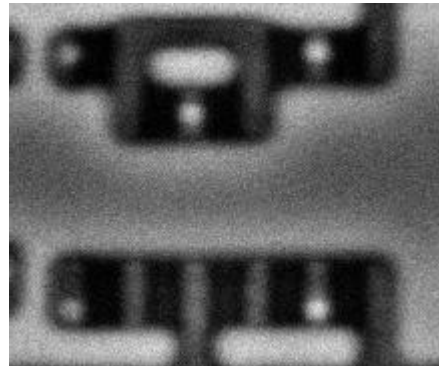
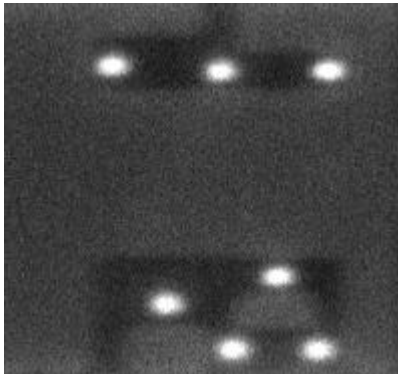
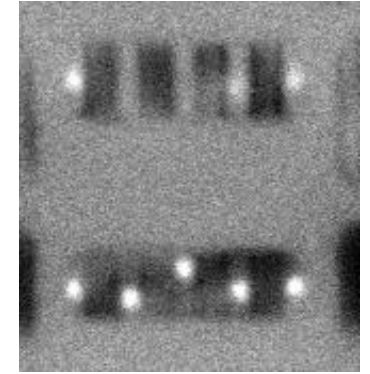
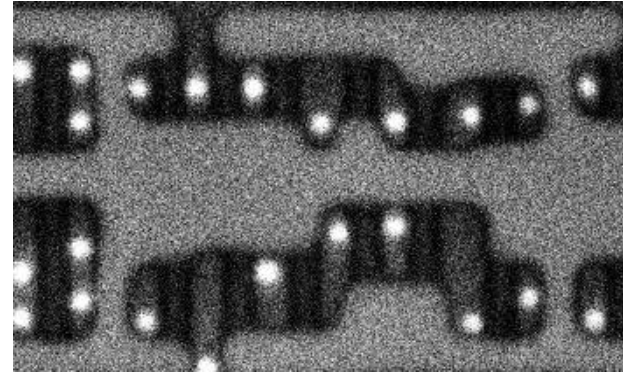
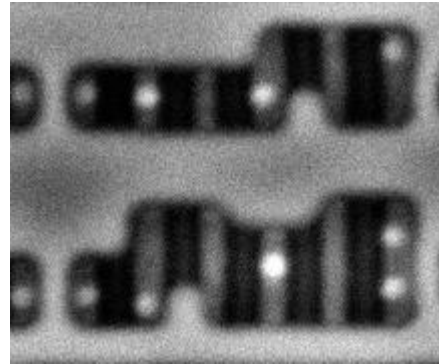
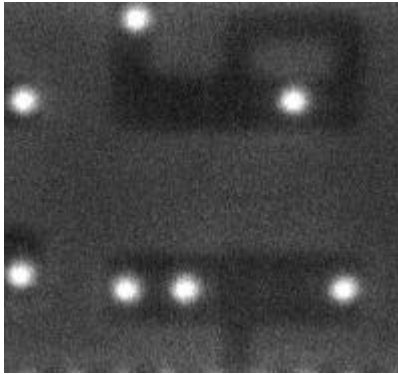
- Use a golden reference model to compare each cell of the same type (= label) against
- If different, count as candidate for modification

TEMPLATE MATCHING VS. SIMILAR CELLS





SELECTED TRUE POSITIVES (↻ REPLACEMENTS)



a) 90nm

b) 65nm

c) 40nm

d) 28nm



DETECTION RESULTS

Chip Statistics

	90 nm	65 nm	40 nm	28 nm
Total Number of Cells	453,850	571,060	917,819	1,467,851
Region of Interest Area	2.089 mm ²	1.848 mm ²	1.052 mm ²	0.962 mm ²

+ Additional Cells	4 / 4 ✓	4 / 4 ✓	4 / 4 ✓	4 / 4 ✓
False Positives	0	0	4	17
↻ Replaced Cells	6 / 6 ✓	6 / 6 ✓	6 / 6 ✓	3 / 6 ✗
False Positives	136	6	11	343

Runtime Effort

Image Acquisition (SEM)	~34 h	~23 h	~53 h	~36 h
Detection Algorithms	~2 h	~3 h	~5 h	~4 h



TROJAN DETECTION: TAKEAWAYS

HARDWARE COMPROMISE IS NOW A PRACTICAL RESEARCH FIELD

Once theoretical, we now have real tooling, test chips, detection workflows, and academic-industry collabs!

HARDWARE SECURITY IS WHERE SOFTWARE WAS 20 YEARS AGO

Once obscure, it's becoming more standard practice, and people have started to care.

DETECTION IS HARD, BUT IT WORKS ...

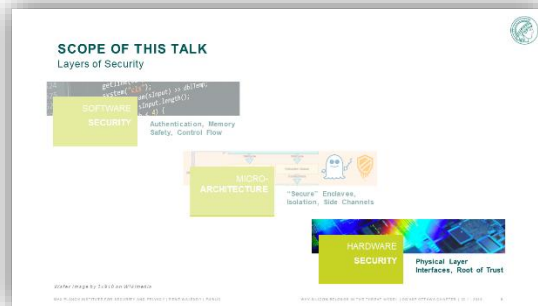
... when done rigorously.
Defensive capability is increasing every year.



WHERE TO GO FROM HERE?

SECURE BOOT != SECURE SILICON

There is more to effective security than soft- and firmware.
Make it part of the threat model.



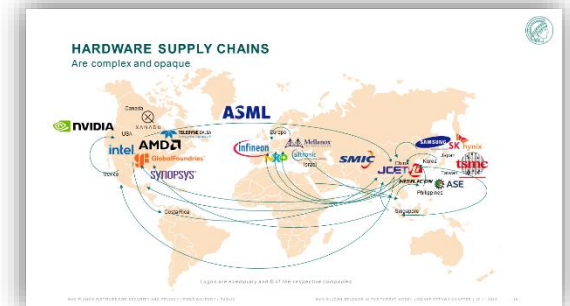
DON'T BLINDLY TRUST HARDWARE

Avoid single points of failure.
Treat hardware like any dependency – consider reputation, patchability, auditability.



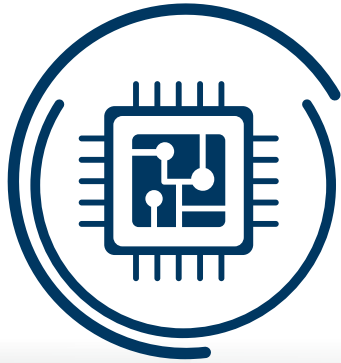
DEMAND TRANSPARENCY

Favor open roots of trust.
In critical environments, push vendors for provenance assurances.





MEET US AT HARRIS 2026



4th HARdware Reverse engineering workShop (HARRIS'26)

March 24-25, 2026
Bochum, Germany

harris2026.mpi-sp.org



Photo © MPI-SP





THANKS FOR YOUR ATTENTION

If you have any questions, feel free to contact me:

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TROJANS IN THE WILD?

... Maybe?

„According to a U.S. defense contractor who spoke on condition of anonymity, a European chip maker recently built into its microprocessors a kill switch that could be accessed remotely.“ [1]

„In this paper we demonstrate how a deliberately inserted backdoor and additional functionalities can be found in the ‘highly secure’ Actel/Microsemi ProASIC3 FPGAs [...] chip used in both military and sensitive industrial applications.“ [2]



[1] „The hunt for the kill switch“ - Sally Adee

[2] „Breakthrough silicon scanning discovers backdoor in military chip“ - Sergei Skorobogatov, Christopher Woods